and *Candida lipolytica* are generally are involved in the production of citric acid.

Citric acid production can be carried out in the following three methods.

a. Koji process or solid state fermentation

b. Liquid surface culture

c. Submerged fermentation

Media used in citric acid production

Citric acid production is carried out by using carbohydrates and n–alkenes. Generally beet molasses, cane molasses, sucrose, commercial glucose and starch hydrolysate are used as carbohydrate sources. The carbohydrate material is diluted and mixed with a nitrogen source (ammonium salts or urea) and the pH and temperature are adjusted according to the process.

Inoculum development

Fungal strains that are used for production are stored in soil or silica gel in the form of spores. Spores are suspended in a freshly prepared sterile water containing between 80 and after a period of growth, it can be used as inoculum for large scale production.

Steps involved in citric acid production

Production Medium

Sucrose, beet molasses, used as carbon source need pretreatment, as it contains excessive amount of trace metals. So ferrocyanide or ferricyanide is added to the production medium before sterilization. Inorganic salts, carbon, hydrogen, oxygen trace metals. Nitrogen, potassium, phosphorus, sulphur and magnesium are taken in Aluminum or stainless steel shallow pans or tray (5–20 cm deep).

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Inoculated with spores of A. *niger* by blowing over the strains of *Aspergillus niger* for fermentation

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The medium is kept at 28–30°C with relative humidity 40–60% and aerated with purified air for 8–12 days

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Citric acid produced is determined by checking the pH or the total acid content of the broth.

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Fermented liquid is drained off and processed further for the recovery of citric acid

Infobits

Influence of trace metals in citric acid production:

Citric acid production is highly influenced by the trace metals. Particularly, iron and manganese in excess amount affect the citric acid production. They affect the cellular morphology and change pellets to filamentous growth (i.e.,) from productive form to unproductive form.

Recovery

The mycelial mat is pressed.

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Milk of lime is added so calcium citrate is formed.

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Again sulphuric acid is added, so calcium sulphate is formed.

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The remaining citric acid solution is filtered and washed. Finally the impure solution of citric acid subjected to treatment with activated carbon and finally pure form of citric acid is collected.

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